NILAM — Simplified System Architecture

Land suitability · ArchAI2 house generation · Kerala construction cost

ONE CONNECTED PRODUCT

LIVE USER FLOW

01 · INPUTS & DATA

User and live services

User input

- Map site or search address
- Radius for land discovery
- Text or English voice request
- House brief and finish level

Live data services

- Google Earth Engine
  - Terrain · land cover · water
- OSM + Kerala hazards
  - Buildings · roads · flood · landslide
- Google location services
  - Maps · Places · Street View

02 · LAND SUITABILITY

Geospatial and multimodal intelligence

Site feature extraction + candidate search

38 mapped features · grid sampling · flat/open-land safety gates

TabPFN 3.5

Tabular site evidence

92% fusion weight

TerraMind

Seasonal Sentinel-2

8% fusion weight

Laya + Whisper

Preference routing

Local speech input

Probability fusion + SHAP explanation

Four classes → final suitability percentage

Land outputs

Score · drivers · ranked flat land · amenities · Street View

03 · HANDOFF

Site context

`SITE_CONTEXT`

Selected site

- Latitude
- Longitude
- District
- Suitability %

Land → home

04 · ARCHAI2

House generation and refinement

Qwen 3.5 9B · local LM Studio

Structured house requirements

Spatial specification

Rooms · floors · setbacks

Adjacency · plot geometry

Layout generation

HouseGAN++ proposals

Procedural optimizer

Deterministic geometry validation

Collision · access · circulation · area · setbacks

Design outputs

Canonical geometry + SQLite revisions

2D plan · 3D model · exports

05 · COST

Kerala estimator

Estimator input

Geometry · district · finish

Cost calculation

- Measured floor area
- Kerala rate bands
- District factor
- Allowances + reserve

Cost outputs

- Budget range
- Cost breakdown
- Source assumptions

OFFLINE PIPELINE

Data preparation, model fitting and evaluation

Public geospatial data

SRTM · CHIRPS · ERA5-Land

Dynamic World · MODIS · JRC

OpenLandMap · hazards · OSM

Statewide training cohort

1,400 points · 100 per district

38 features + weak labels

Paired dry / monsoon image chips

Model fitting

TabPFN in-context fit

TerraMind fine-tuning

District-based data split

Calibration + evaluation

92% TabPFN + 8% TerraMind

Accuracy · precision · recall

Macro F1 · log loss · ablations

Versioned runtime artifacts

TabPFN estimator · TerraMind checkpoint

Fusion configuration · feature schema

Metrics and model metadata

React + Google Maps UI · Nilam FastAPI · ArchAI2 FastAPI · Next.js workspace · Local LM Studio · SQLite + local caches